

# Data Structures

SITAMS CSE (AI & ML) • R23 • I Year • II Semester • Course Code: 23CSE121T

**Source basis:** Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

## Unit 1: Introduction to Linear Data Structures

### Syllabus topics

- Definition and importance of linear data structures
- abstract data types (ADTs) and their implementation
- overview of time and space complexity for linear data structures
- linear and binary searching
- bubble, selection and insertion sorting

### Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

## Unit 2: Linked Lists

### Syllabus topics

- Singly linked lists: representation and operations
- doubly linked lists and circular linked lists
- comparing arrays and linked lists
- applications of linked lists

### Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

## Unit 3: Stacks

### Syllabus topics

- Introduction to stacks
- properties and operations
- implementation using arrays and linked lists
- applications in expression evaluation, backtracking and reversing lists

### Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.

- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

## Unit 4: Queues and Deques

### Syllabus topics

- Introduction to queues
- properties and operations
- implementation using arrays and linked lists
- applications in breadth-first search and scheduling
- deques and their applications

### Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

## Unit 5: Trees and Hashing

### Syllabus topics

- Introduction to trees
- binary search tree insertion, deletion and traversal
- hashing and hash functions
- collision resolution using chaining and open addressing
- hash tables and applications

### Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

### Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.